

Seismic Fragility Analysis of Residential RC Buildings in Kathmandu Valley Built Before the Gorkha Earthquake 2015 with Retrofitting Interventions

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ABSTRACT

Kathmandu Valley is highly vulnerable to seismic events due to its geological location, dense population, and rapid urbanization. Typical residential buildings inside the valley are low-rise Reinforced Concrete (RC) structures, and those constructed before the 2015 Gorkha earthquake are mostly built without following seismic codes. Previous studies assessed the seismic vulnerability of RC buildings in Kathmandu Valley. Still, they were limited to a small number of buildings and did not account for different plan configurations, making generalization difficult. This study assessed 12 existing residential RC buildings in Kathmandu Valley using fast non-linear dynamic analysis to evaluate their seismic vulnerability. Retrofitting interventions, including RC jacketing of columns and concrete jacketing for beams, were applied to enhance structural performance and ensure compliance with the Nepalese building codes. The adopted retrofitting technique significantly improved the overall seismic performance of the buildings, reducing the mean exceedance probabilities of the Immediate Occupancy (IO), Life Safety (LS), and Collapse Prevention (CP) damage states by approximately 78%, 90%, and 95%, respectively. Furthermore, lower-rise buildings such as BD2, BD3, BD5, BD7, BD8, and BD11 exhibited particularly notable improvements, with LS exceedance probabilities reducing by approximately 89%-98% following retrofitting. The research concluded that the adopted jacketing retrofitting intervention works exceptionally well for buildings with lesser heights, but as the height increases, the effectiveness of this method decreases, highlighting the need for more extensive retrofitting strategies for taller structures.

Keywords: Kathmandu valley; Seismic Vulnerability; Reinforced Concrete Structure; Fast Non-Linear Dynamic Analysis; Retrofitting

1. Introduction

An earthquake is a long-term complex stress accumulation and release process occurring in a highly inhomogeneous medium of the Earth's interior [1]. It is estimated that each year, there are around 500,000 detectable earthquakes in the world, of which 100,000 can be felt and about 100 cause damage [2]. Most of these earthquakes are induced by natural factors like tectonic plate movement, volcanism, etc., while some are induced by artificial factors such as deep mining, large reservoirs, and nuclear explosions. Each year, earthquakes cause significant structural damage and loss of life worldwide.

Nepal is among the most seismically active regions globally, ranking 11th worldwide in terms of seismic risk [3], [4]. The high seismic activity in this region is caused by active tectonic processes, where the Indian plate is being subducted beneath the Eurasian plate. Many great earthquakes have been recorded in this region, among which the 1934 Bihar–Nepal earthquake (Mw 8.2) and the 2015 Gorkha earthquake (Mw 7.8) were the most devastating in the past century [5]. During both of the mentioned devastating earthquakes, the Kathmandu Valley was the hardest hit. The highest number of fatalities and structural collapses were recorded in this area [6]. The reason for this recurring destruction during major earthquakes in the Kathmandu Valley is its soft alluvial and lacustrine soil, and as the capital of the country is within this area, the population is also dense in this part. A simple example of how massive the destruction was in the Kathmandu Valley is that nearly 12,500 houses were destroyed during the 1934 Bihar–Nepal earthquake, which was around 19% of the total number of houses in the valley at that time [7]. Similarly, during the 2015 Gorkha Earthquake, Kathmandu Valley witnessed severe destruction, where 103,022 private houses were destroyed, including both reinforced concrete and unreinforced masonry structures [8]. The reason for the failure of reinforced concrete buildings in the Kathmandu Valley during the Gorkha earthquake was mainly improper construction practice, lack of proper structural design and detailing, and construction of buildings without proper soil exploration [9].

Old residential RC buildings in the Kathmandu Valley pose a significant risk of damage during earthquakes, as most structures do not comply with current building codes. These buildings exhibit structural deficiencies such as asymmetric plan configurations, mass irregularities, and inadequately sized beams and columns. To determine appropriate retrofitting solutions to mitigate the adverse effects of these structural deficiencies, seismic fragility assessment of buildings is essential. Seismic fragility assessment is a probabilistic approach used to estimate the likelihood of exceeding a specified damage level at a given level of seismic intensity. Many researchers have adopted this approach to study the vulnerability of structures under different types and levels of seismic loading. For example, Kirçil & Polat studied the fragility of 3 mid-rise RC frame buildings in Istanbul, which were designed using the 1975 Turkish Code [10]. Using Incremental Dynamic Analysis, they introduced the concepts of combined fragility and extended fragility. Combined fragility curves were developed in this study to demonstrate the unified response of 2 different reinforcement grades. Extended fragility curves were developed using regression analysis, which were used to determine the relationship between the fragility curve parameters and the number of storeys. With the help of an extended fragility curve, it was observed that fragility parameters change significantly due to the number of storeys. Another study done in Taiwan by Liao et al. developed fragility functions for various building types in Taiwan within the Taiwan Earthquake Loss Estimation System (TELES), adapting the HAZUS framework to local conditions [11]. Buildings were classified into 15 categories. The classification was based on structural characteristics like material type and height, with design levels (High-, Moderate-, Low-Code) reflecting the construction era. Using nonlinear static pushover analysis, fragility curves were derived for different damage states based on peak ground acceleration (PGA) and validated against observed damage from the 1999 Chi–Chi Earthquake. This study provides essential parameters for seismic risk and loss estimation. Also, one study in South America conducted by Villar-Vega et al. reviewed the different existing fragility models in that region with a focus on the Andean region [12]. They developed a unified fragility model for the residential building stock for that region, where seismic vulnerability is particularly high due to rapid urbanization and non-engineered construction. The study employed nonlinear time history analysis of equivalent single-degree-of-freedom systems under recorded ground motions to derive damage distributions and corresponding fragility curves for representative fifty-four building classes.

In Nepal, many studies have also been conducted on a similar topic. The study conducted by Budthapa et al. used Incremental Dynamic Analysis to investigate the impact of irregular mass distribution on a symmetrical RCC building by the addition of a heavy mass structure in different locations of the 9 and 12-storey building and concluded that irregular mass distribution at higher levels is more vulnerable to damage during an earthquake than mass concentration at lower levels [13]. Also, this study determined that mass-heavy structures at lower levels have better performance during earthquakes than regular structures. Another research done by Rodrigues et al. conducted a seismic vulnerability assessment of a reinforced concrete school building in Kathmandu [14]. Using a calibrated 3D non-linear numerical model, the study revealed that irregular infill distribution and column weaknesses contributed to soft-storey and torsional mechanisms, leading to significant drift demands and failure risks, especially at the first and third storeys. The authors tested four retrofit strategies-introduction of infill masonry walls, partial and full RC column jacketing, and steel bracing-and compared their effectiveness through non-linear static and dynamic analyses. Among these, full RC column jacketing proved to be the most effective in reducing inter-storey drifts and enhancing overall stiffness, while steel bracing and partial interventions showed limited or adverse effects. The study highlights the high seismic vulnerability of typical Nepali school buildings and underscores the urgent need for comprehensive retrofitting strategies, with RC jacketing emerging as a viable and efficient solution. Kc & Maskey analyzed regular ordinary moments resisting RC framed structures of 4 and 7 storeys inside Kathmandu valley using Incremental Dynamic Analysis, where short and long duration ground motions were used, and concluded that the duration of the earthquake has a significant impact on the seismic performance of the buildings [15]. The longer the duration of the earthquake, the greater the effect on the structures, especially in the case of taller buildings. In another research, Adhikari & Maskey investigated the seismic performance of the RC residential buildings with different plan configurations using the pushover analysis method and non-linear time history analysis method [16]. The findings showed that for the structures of the same storey configurations, the probability of failure at the same damage state differs with respect to the plan configurations of buildings, in which the asymmetric buildings showed more vulnerability due to irregularity in mass and stiffness.

The study by Aryal et al. investigated the seismic vulnerability of 6 low-rise reinforced concrete buildings situated in Lalitpur Metropolitan City (LMC) [17]. After designing and comparing the buildings by non-engineered, pre-engineered, and engineered design methods and using nonlinear static and dynamic analyses in OpenSees, the authors demonstrated that non-engineered and pre-engineered buildings are significantly more susceptible to earthquake damage, while retrofitting interventions such as RC jacketing, steel jacketing, and FRCM(Glass and Hemp) jacketing systems substantially enhanced the stiffness, strength, and overall seismic performance of those buildings. The study also employed fragility analysis to evaluate the probability of different damage states under varying levels of seismic intensity, allowing a probabilistic assessment of structural performance and retrofit effectiveness. This approach was used to quantify how retrofitting reduces the likelihood of exceeding critical performance levels such as operational life safety and collapse prevention limit states. Among the retrofit techniques studied, Glass-FRCM was found to be the most effective, as higher base shear was achieved by adopting this method. Ghimire & Chaulagain studied the behavior of the 3 Blocks of the reinforced concrete Building of Pokhara University using non-linear static and linear dynamic analysis. The study revealed that an increase in PGA leads to an increase in fragility, and also the fragility curve of a building depends upon building properties and configuration rather than on earthquake data [18]. In one research, Suwal & Uprety conducted a seismic fragility assessment of a low-rise RC frame where structures with and without the unreinforced masonry infill walls were compared [19]. Infill walls were modelled using a macro modelling approach. The comparison revealed that at a seismic demand of 0.8g, the probability of exceedance of corresponding damage states was significantly higher in the case of bare frame than in the infill case. This study also suggested the inclusion of the infill effect during the design of low-rise RC frame structures in the context of Nepal.

Poudel & Shrestha assessed the seismic vulnerability of a code non-compliant mixed masonry-RC building in Panauti, Nepal, through nonlinear pushover analysis and fragility assessment of four splint and bandage retrofitting configurations using DIANA FEA [20]. Fragility analysis revealed that the unretrofitted building exhibited nearly 100% probability of exceeding both Immediate Occupancy and Life Safety limit states, which was dropped

significantly in the fully retrofitted model. While the normal splint and bandage retrofitting configuration offered the highest strengthening with 85% base shear enhancement and also 0% probability of exceedance for all limit states, the diagonal splint and bandage was found to be an optimal retrofitting solution providing a balance of strength economy and ductility. This study concluded that the ferrocement splint and bandage is a cost-effective strengthening strategy for seismically vulnerable masonry buildings in resource-constrained regions. Research conducted by Gupta et al. investigated the seismic strengthening of a representative non-engineered RC building using vegetal FRCM retrofitting through numerical modeling [21]. The study considered four organic fibers: hemp, flax, sisal, and cotton for retrofitting, applied in 1-ply and 2-ply configurations. The authors first validated numerical models of hemp-strengthened beams against experimental and analytical results from other researchers. Then, the representative non-engineered RC building was strengthened using these four organic fibers. Nonlinear pushover analysis showed that retrofitted buildings exhibited increased base shear capacity. For 1-ply systems, flax provided the greatest improvement in base shear, whereas for 2-ply systems, hemp showed the most significant improvement in base shear compared to the unretrofitted model. Fragility analysis further demonstrated a reduction in seismic vulnerability. Hemp was the most effective among 1-ply FRCM retrofits, with almost zero probability of exceedance in the CP state. In the case of 2-ply FRCM retrofits, flax had the lowest probability of exceeding all three limit states, confirming the potential of vegetal FRCM as a sustainable retrofitting technique for seismically deficient buildings.

While most previous studies in Nepal have assessed the vulnerability of RC buildings, they were limited to a relatively small sample size, highlighting the need for research that incorporates a larger number of buildings. Moreover, these studies did not account for variations in plan configurations, which limited the generalizability of their findings. In this study, 12 existing residential RC buildings within the Kathmandu valley, built before the 2015 Gorkha Earthquake, ranging from 3 to 5 storeys with varying plan configurations, were first modeled in ETABS V21 software. Fragility curves for 3 damage states, namely Immediate Occupancy, Life Safety, and Collapse Prevention, with PGA as intensity measure, are drawn for each building, and the fragility of the building is assessed. Later in this study, reinforced concrete column jacketing in columns and concrete jacketing in beams was adopted for all buildings in order to make them adhere to the current national codes, and the fragility curve after retrofitting was drawn and compared with the fragility curve before retrofitting. With the help of 12 representative buildings, the overall vulnerability of RCC residential buildings of Kathmandu valley built before the 2015 Gorkha Earthquake was assessed.

2. Methodology

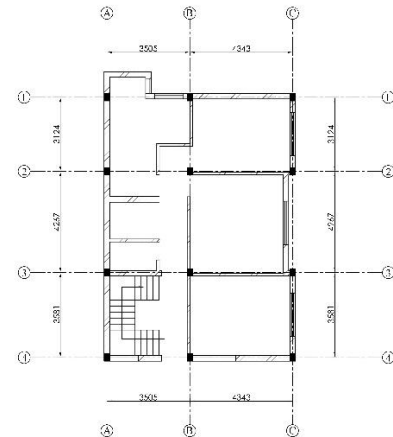
2.1. Building description

Twelve residential RCC buildings, ranging from 3 to 5 storeys, were selected. Their actual photographs, along with their as-built drawings, are shown in Figure 1. Buildings within this storey range were chosen because they represent the most common residential RCC structures in the Valley. The selected buildings have different plan configurations to account for the effects of geometric irregularity. Table 1 provides a description of the buildings, including their total height, number of storeys, column and beam dimensions, and column reinforcement details. The slab size for each building model is taken as 125mm.

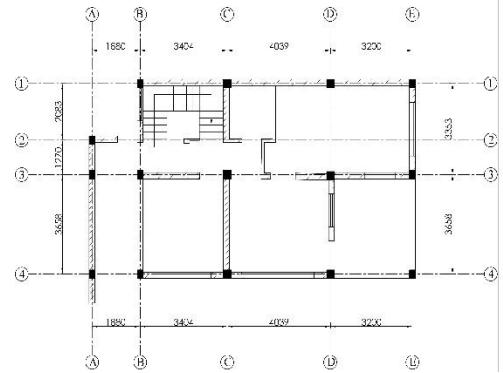
Table 1: Building Description With their storey, column size, and beam size data

Building Name	No. of Storey	Total Height(m)	Beam Size (mm)	Column Size (mm)	Column Reinforcement
BD 1	5	14.22	230×230	230×305	4-16 ϕ

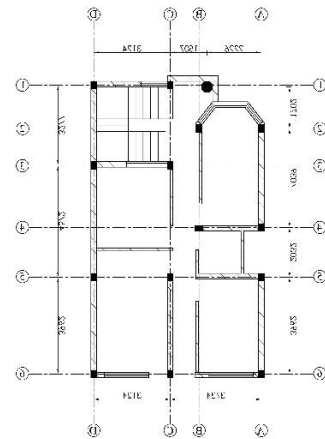
BD 2	4	10.97	230×230	230×305	4-16 ϕ
BD 3	4	10.97	230×230	230×305	4-16 ϕ
BD 4	4	11.38	230×230	230×305	4-16 ϕ
BD 5	3	8.53	230×230	230×305	4-16 ϕ
BD 6	4	11.38	230×230	230×230 230×305	4-16 ϕ
BD 7	3	8.84	230×230	230×305	4-16 ϕ
BD 8	3	8.23	230×230	230×230	4-16 ϕ
BD 9	5	14.22	230×230	230×230	4-16 ϕ
BD 10	4	11.38	230×230	230×305	4-16 ϕ
BD 11	3	8.53	230×230	230×305	4-16 ϕ
BD 12	5	14.22	230×230	230×230	4-16 ϕ



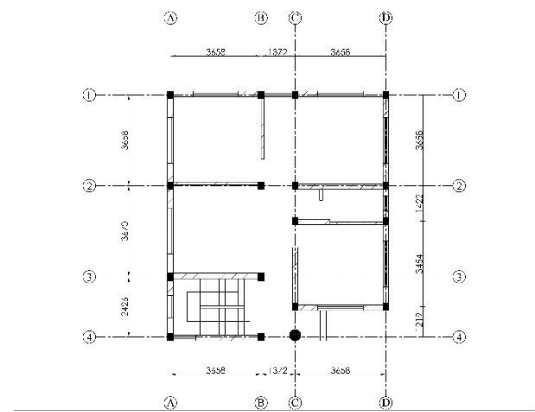
a. BD 1



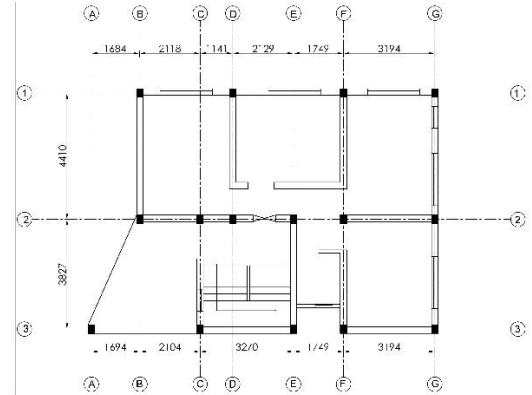
b. BD 2



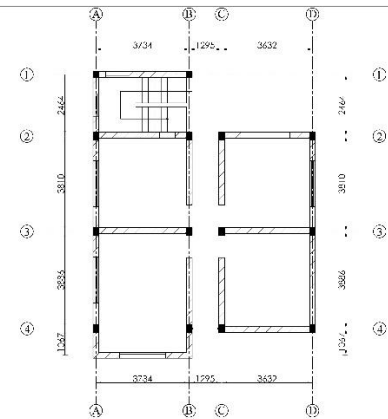
c. BD 3



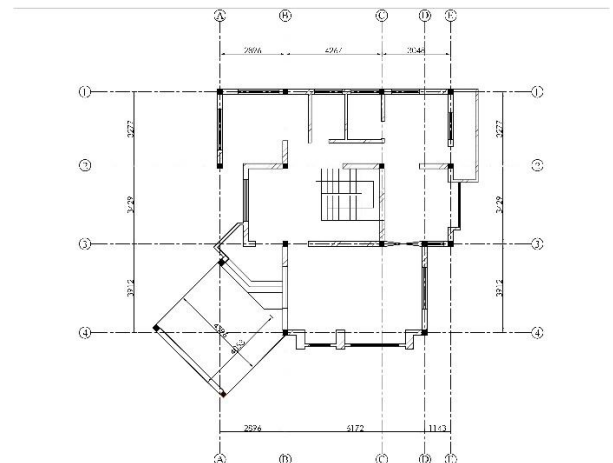
d. BD 4



e. BD 5

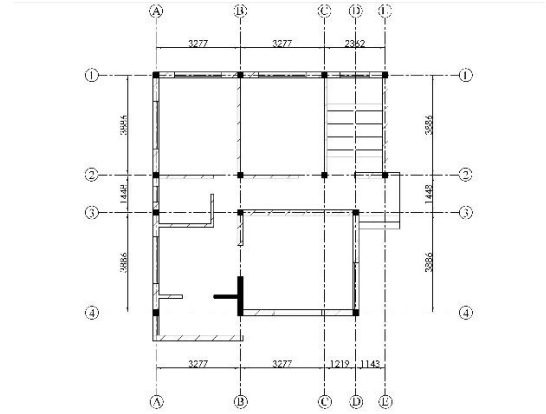


f. BD 6

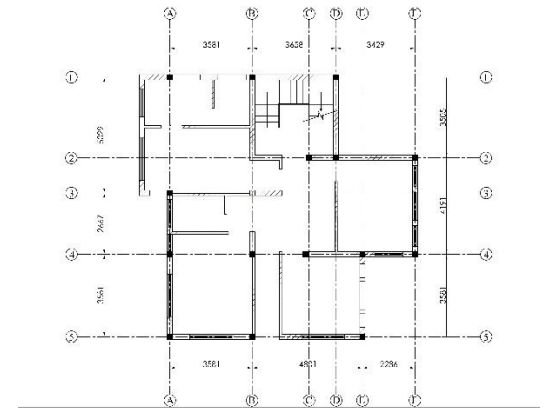


i.

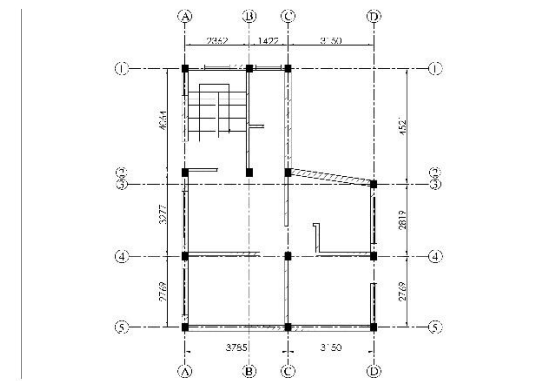
g. BD 7



h. BD 8



i. BD 9



j. BD 10

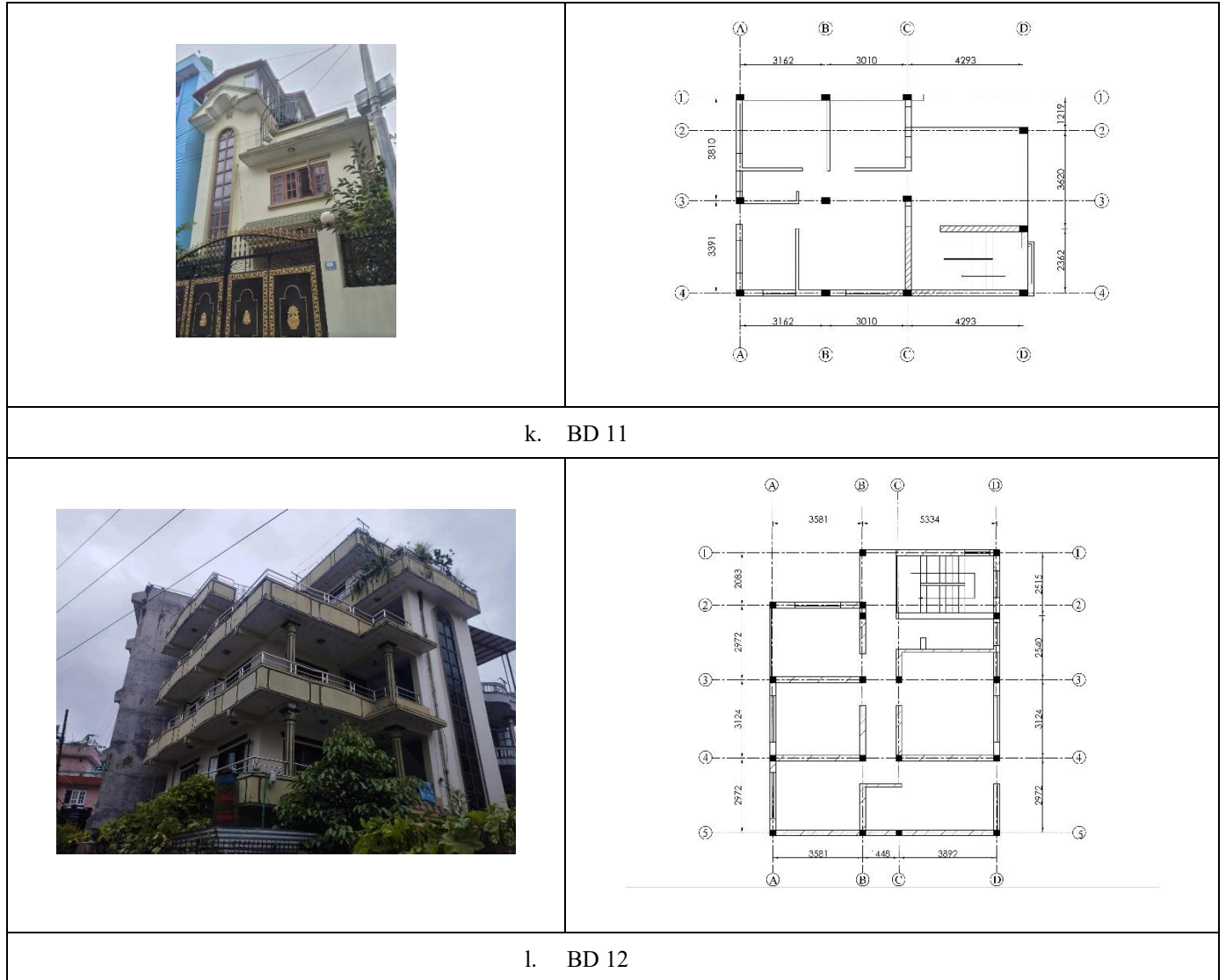


Fig.1 Buildings and Their Ground Floor Plans Constructed Before 2015 Selected for Study

2.2 Material and load description

Concrete grade of M20 and rebar of Fe 415 were used for all building models. Infill walls were not modeled directly, but rather applied as loads, and the unit weight of brick masonry was taken as 18.85 kN/m^3 . Similarly, the unit weight of concrete was taken as 25 kN/m^3 following IS 875 Part I [22]. Live and dead load intensities were adopted following IS 875 Part I [22] and Part II [23], with live loads of 2 kN/m^2 for rooms, kitchens, toilets, and bathrooms; 3 kN/m^2 for balconies, terraces, staircases, and passages; zero roof live load; and a floor finish load of 1 kN/m^2 . The material degradation and stiffness degradation were accounted for by adopting suitable modifiers as mentioned in the seismic design code of Nepal, NBC 105:2025 [24].

2.3 Non-linear modelling

The nonlinear behavior of structures primarily arises from geometric and material nonlinearities. In this study, geometric nonlinearity was incorporated using the P-Delta analysis option available in ETABS. The Iterative-Based

on Load method was adopted to evaluate P-Delta effects, wherein an iterative solution procedure is employed to account for second-order effects.

To model material nonlinearity, P-M2-M3 auto hinges were assigned to columns in accordance with ASCE 41-17, Table 10-8. These hinges were placed at relative locations of 0.1 and 0.9 along the column length. Similarly, M3 auto hinges were assigned to beams based on ASCE 41-17, and were also located at relative positions of 0.1 and 0.9 from the member ends. These hinge properties were used to simulate the inelastic behavior of structural members during nonlinear analysis [25].

To capture the cyclic behavior and energy dissipation during earthquakes, the steel reinforcements were modeled using the kinematic hysteresis model, which is based on the principle of kinematic hardening of metals. The kinematic hysteresis model, available as one of the standard cyclic models in ETABS, is well-suited for ductile materials [26]. For concrete members, the Takeda hysteresis model was adopted. This model produces a pinched and degrading hysteresis loop, effectively representing the stiffness and strength degradation observed in reinforced concrete under cyclic loading [27],[28].

2.4 Ground motion information

A total of 10 ground motion records were taken for this study. The ground motion records were selected in such a way that they possess unique characteristics in themselves with variation in magnitude, duration (short, long, and moderate), and frequency (low, moderate, and high). The selected ground motion records are a combination of both near-field and far-field earthquake shaking. Seven earthquakes are near fields, and the remaining three are far-field earthquakes. The table containing information on selected ground motions is given in Table 2. Fast Nonlinear Analysis (FNA) was performed using the selected 10 earthquake ground motion records. The records were incrementally scaled in steps of 0.1g based on their peak ground acceleration (PGA) values to conduct the Incremental Dynamic Analysis (IDA). Figure 3 illustrates the response spectra of the 10 selected ground motion records.

Table 2: List and Characteristics of Selected Earthquakes

Name	Year	Mw	PGA	Near/Far Field	Epicenter
Comendocino	1992	7.1	0.637g	Near Field	On-shore 4km west of Petrolia
Erzincan	1992	6.7	0.432g	Near Field	7.7 km from the eastern end of the North Anatolian Fault
Gorkha	2015	7.8	0.177g	Far-Field	Barpak, Gorkha, Nepal
Kobe	1995	6.9	0.833g	Near-Field	Awaji Island, Japan
Landers	1992	7.3	0.713g	Far-Field	Near the town of Landers in Southern California, close to the Johnson Valley Fault
Loma Prieta Lex dam	1989	6.9	0.686g	Near Field	Loma Prieta Peak in the Santa Cruz Mountains, USA
Loma Prieta Los Gatos	1989	6.9	0.717g	Far field	Loma Prieta Peak in the Santa Cruz Mountains, USA
Northridge Olive View	1994	6.7	0.732g	Near Field	along a "blind" thrust fault, beneath the San Fernando Valley

Northridge Rinaldi	1994	6.7	0.889g	Near Field	along a "blind" thrust fault, beneath the San Fernando Valley
Tabas	1978	7.4	0.899g	Near Field	near Tabas city, located close to the Kahak Fault zone

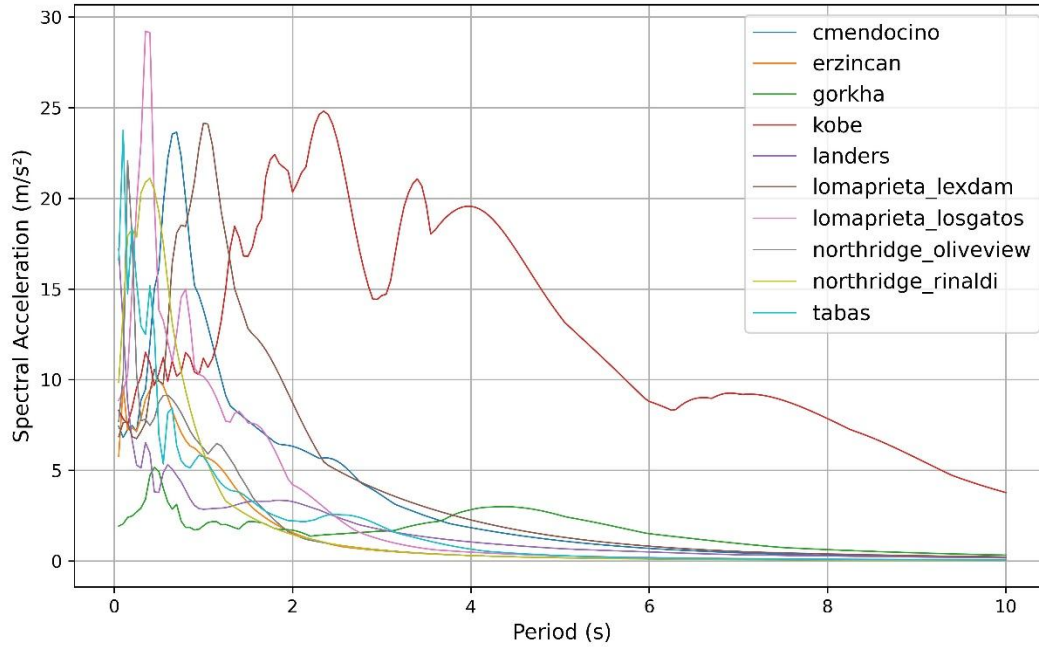


Fig.2 Response Spectrum of Selected Earthquakes

3. Result and Discussion

3.1 Fragility Analysis and Fragility Curves

The fragility of a structure represents the probability that the structure will exceed a specified damage state for a given value of an intensity measure. Immediate Occupancy (IO), Life Safety (LS), and Collapse Prevention (CP) were the limit states taken in this study. Similarly, the inter-storey drift ratio (%) was taken as the damage measure, and Peak Ground Acceleration (PGA) was taken as the Intensity measure. The limiting values of inter-storey drift (%) for IO, LS, and CP, according to FEMA (Federal Emergency Management Agency) [29], are given in Table 3.

Table 3: Inter Storey Drift Ratio (%) Limits [29]

Limit State	Inter Storey drift ratio (%)
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Immediate Occupancy (IO)	1
Life Safety (LS)	2
Collapse Prevention (CP)	4

By scaling each ground motion record with respect to the individual PGA with an increment of 0.1g, corresponding drift values were obtained for 10 different ground motions. After that, for each damage state, 10 corresponding PGA values were noted from the incremental analysis results, whose scatter plot is presented in Figure 4. Fragility curves were then drawn based on the obtained mean and standard deviation of $\ln(\text{PGA})$ values, where the x-axis represents the drift % and the y-axis represents the cumulative probability of occurrence of damage, equal to or greater than damage level D [10], which is expressed as:

$$P(\leq D) = \Phi\left\{\frac{\ln X - \lambda}{\beta}\right\} \text{ where,}$$

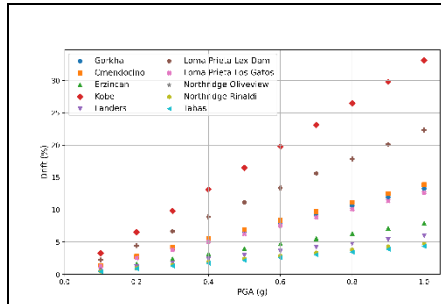
X = PGA values, Φ = standard normal distribution, λ = mean of $\ln X$, β = standard deviation of $\ln X$

The corresponding values of λ and β of 12 different buildings before and after retrofitting for each damage state is shown in Table 4:

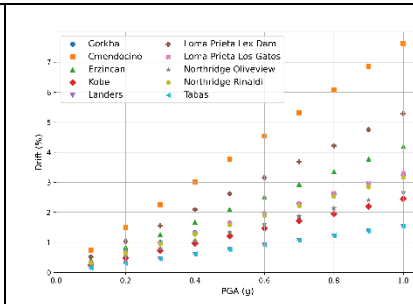
Table 4: Mean and Standard Deviation Values for different limit states before and after retrofitting

Building Name	Immediate Occupancy				Life Safety				Collapse Prevention			
	λ		β		λ		β		λ		β	
	Before	After	Before	After	Before	After	Before	After	Before	After	Before	After
BD 1	-2.257	-1.455	0.680	0.564	-1.573	-0.769	0.694	0.572	-0.885	-0.080	0.700	0.576
BD 2	-1.131	-0.280	0.490	0.462	-0.442	0.412	0.497	0.462	0.249	1.104	0.501	0.462
BD 3	-2.061	-0.874	0.622	0.577	-1.382	-0.187	0.627	0.579	-0.696	0.504	0.631	0.580
BD 4	-1.467	-0.343	0.538	0.463	-0.773	0.350	0.538	0.462	-0.082	1.043	0.538	0.462
BD 5	-1.139	-0.124	0.510	0.495	-0.446	0.569	0.511	0.495	0.246	1.262	0.511	0.496
BD 6	-1.801	-0.733	0.596	0.543	-1.120	-0.047	0.602	0.546	-0.433	0.643	0.605	0.548
BD 7	-1.008	0.006	0.461	0.425	-0.319	0.637	0.466	0.491	0.372	1.278	0.469	0.589
BD 8	-1.170	0.244	0.492	0.397	-0.481	0.700	0.493	0.400	0.210	0.141	0.494	0.403
BD 9	-1.876	-0.446	0.610	0.509	-1.190	0.245	0.616	0.511	-0.501	0.937	0.619	0.512
BD 10	-1.623	-0.271	0.552	0.454	-0.939	0.403	0.558	0.450	-0.250	1.086	0.561	0.449

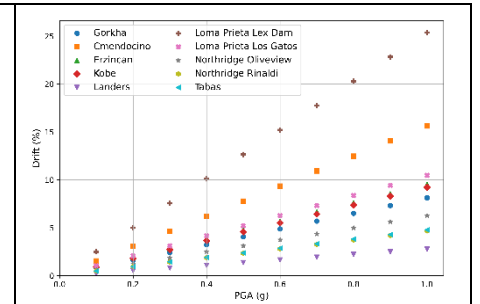
BD 11	-1.363	0.125	0.458	0.427	-0.670	0.818	0.458	0.427	0.023	1.511	0.458	0.427
BD 12	-1.995	-0.542	0.589	0.531	-1.310	0.145	0.594	0.534	-0.621	0.834	0.597	0.536



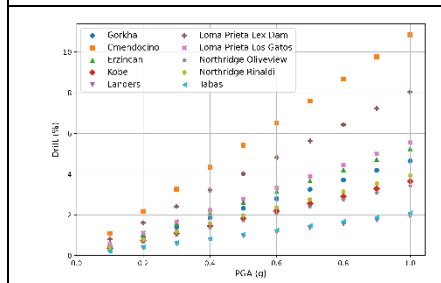
(a) BD 1



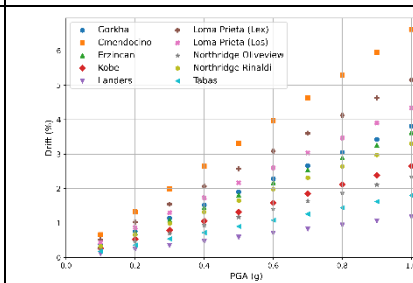
(b) BD 2



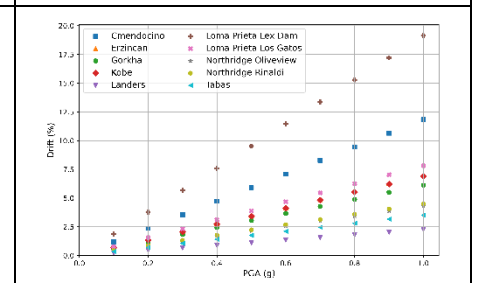
(c) BD 3



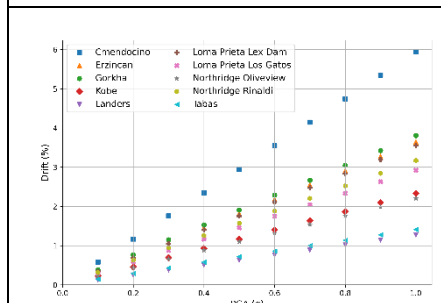
(d) BD 4



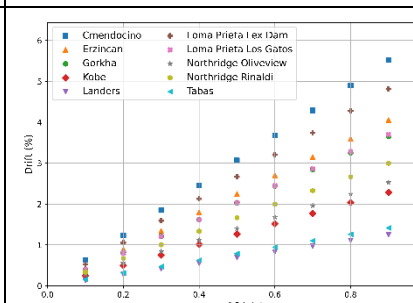
(e) BD 5



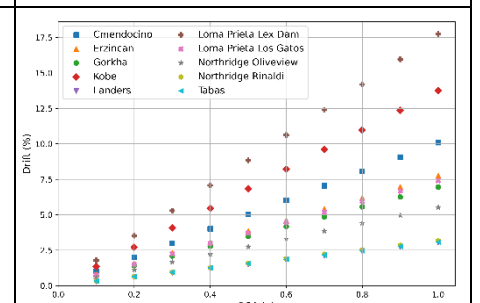
(f) BD 6



(g) BD 7



(h) BD 8



(i) BD 9

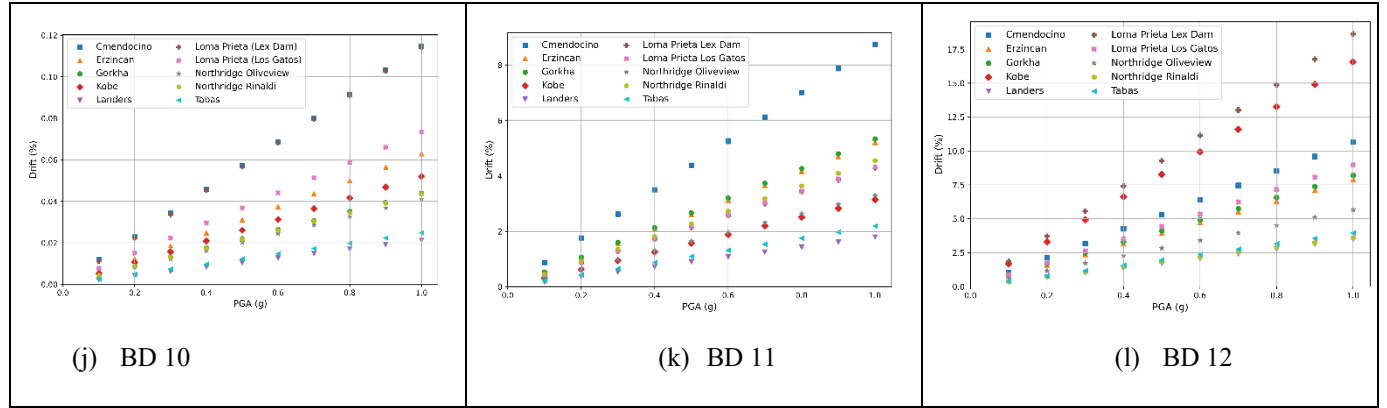


Fig.3 Scatter IDA plots of Selected Ground Motions

3.2 Retrofitting Technique

Nepal retrofit guidelines published after the 2015 Gorkha earthquake specifically recommend reinforced concrete (RC) jacketing as a standard method for strengthening RC structural members such as beams and columns [30]. Also, the study done by Rodrigues et al. concluded that RC column jacketing is the most effective retrofitting technique for RC structures in Nepal, where the comparison was made among four retrofitting techniques to determine their effectiveness [14]. This is because, in building frames, beams and columns play a key role in absorbing earthquake forces, and during an earthquake, the movement of columns helps to dissipate the energy. As a result, repairing or strengthening columns and beams becomes necessary to make sure the structure can function properly after the earthquake [31]. Reinforced concrete jacketing improves strength, stiffness, and ductility, which are critical for earthquake resistance. RC jacketing retrofitting technique was applied by following IS 15988:2013 [32].

The amount of steel and concrete required was provided in accordance with IS 15988:2013 [32], where the amount of concrete and steel was increased by 1.5 and 1.33 times the amount of concrete and steel required, respectively. The reason for increasing the amount of both concrete and steel than required was to account for the losses during construction. The existing section of columns was insufficient according to the results obtained from analysis using the equivalent static method. The existing beams of 230 x 230 mm size, they passed the design check, but concrete jacketing was still adopted for the beams to meet the minimum size criteria mentioned in the prevalent building codes of Nepal. The minimum size of columns and beams was adopted by following mainly two codes: 1.) NBC 105:2025 [24], which is the code for seismic design of buildings in Nepal, 2.) NBC 205:2024 [33], which is ready to use detailing guidelines for low-rise reinforced concrete buildings.

After retrofitting through concrete jacketing, the beam dimensions for all buildings have been increased and adapted to 250 x 355 mm. The reason for adopting this size for all beams is that it is the minimum size recommended in NBC 205: 2024 [30]. In this study, the amount of reinforcement and section size required for columns were designed and analyzed from the Equivalent Static Method, where building importance factor was taken as 1 for ordinary structure, seismic zone factor 0.35, and soil type D for Kathmandu valley. The ductility factor for SLS (Serviceability Limit State) and ULS (Ultimate Limit State) was taken as 1 and 4, whereas the overstrength factor was taken as 1.25 and 1.5, respectively. Similarly, the time period of buildings was obtained from the respective ETABS model before and after retrofitting.

Table 5: Section and the time period details of the buildings after retrofitting

Building ID	Column Size		Beam size before retrofitting	Beam size after retrofitting	Time period before retrofitting (sec)	Time period after retrofitting (sec)	Column reinforcement		
	Before retrofitting (mm)	After retrofitting (mm)					Before retrofitting	After retrofitting	Added reinforcement
BD 1	230×305	420×420	230x230	355x250	1.729	0.942	4-16 ϕ	16-12 ϕ & 4-16 ϕ	16-12 ϕ
BD 2	230×305	400×400	230x230	355x250	0.934	0.428	4-16 ϕ	16-12 ϕ & 4-16 ϕ	16-12 ϕ
BD 3	230×305	350×350	230x230	355x250	1.087	0.496	4-16 ϕ	16-12 ϕ & 4-16 ϕ	16-12 ϕ
BD 4	230×305	350×350	230x230	355x250	0.751	0.378	4-16 ϕ	16-12 ϕ & 4-16 ϕ	16-12 ϕ
BD 5	230×305	350x350	230x230	355x250	1.298	0.689	4-16 ϕ	16-12 ϕ & 4-16 ϕ	16-12 ϕ
BD 6	230×230 & 230×305	350×350	230x230	355x250	1.201	0.522	4-16 ϕ	12-12 ϕ & 4-16 ϕ	12-12 ϕ
BD 7	230×305	350×350	230x230	355x250	0.642	0.353	4-16 ϕ	8-12 ϕ & 4-16 ϕ	8-12 ϕ
BD 8	230×230	350×350	230x230	355x250	0.712	0.307	4-16 ϕ	8-12 ϕ & 4-16 ϕ	8-12 ϕ
BD 9	230×230	420×420	230x230	355x250	1.326	0.547	4-16 ϕ	16-12 ϕ & 4-16 ϕ	16-12 ϕ
BD 10	230×305	400×400	230x230	355x250	0.838	0.367	4-16 ϕ	16-12 ϕ & 4-16 ϕ	16-12 ϕ
BD 11	230×305	350×350	230x230	355x250	0.589	0.303	4-16 ϕ	16-12 ϕ & 4-16 ϕ	16-12 ϕ
BD 12	230×230	400×400	230x230	355x250	1.374	0.539	4-16 ϕ	16-12 ϕ & 4-16 ϕ	16-12

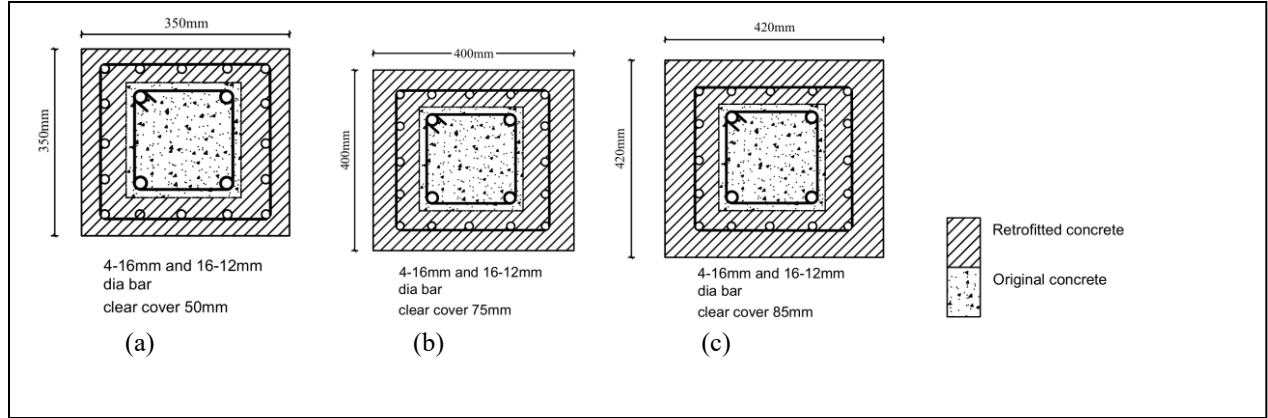


Fig.4 Cross Section of Column after Retrofitting

Table 5 summarizes the column section properties, reinforcement details, and corresponding changes in the fundamental time period of the buildings before and after retrofitting. Before retrofitting, most buildings, BD 1 to BD 7, BD 10, and BD 11 had column sizes of 230×305 mm, while Buildings BD 8, BD 9, and BD 12 had smaller sections of 230×230 mm. All buildings were originally reinforced with 4 longitudinal bars of 16 mm diameter, which were found to be inadequate for the given seismic demand. After retrofitting, the column sizes were increased to 350×350 mm Figure 5(a), 400×400 mm Figure 5(b), or 420×420 mm Figure 5(c), depending on the level of deficiency observed in each building. Buildings BD 1 and BD 9 were retrofitted with larger column sizes of 420×420 mm, while BD 2, BD 10, and BD 12 adopted 400×400 mm sections. The remaining buildings were strengthened using 350×350 mm columns. In terms of longitudinal reinforcement, the retrofitting involved the addition of new reinforcing bars while retaining the existing reinforcement. For Buildings BD 1, BD 2, BD 3, BD 4, BD 5, BD 9, BD 10, BD 11, and BD 12, an additional 16–12 Φ bars were provided, resulting in a combined reinforcement of 16–12 Φ and 4–16 Φ after retrofitting. Buildings BD 6 received 12–12 Φ additional bars, while BD 7 and BD 8 were strengthened with 8–12 Φ additional bars, reflecting relatively lower strengthening demand, which is shown in Table 5. The enhancement in size and reinforcement led to a significant reduction in the fundamental time period of all buildings. The time period before retrofitting ranged from 0.589 s to 1.729 s, which reduced to values between 0.303 s and 0.942 s after retrofitting. This reduction clearly indicates an increase in lateral stiffness and improved seismic performance of the retrofitted buildings.

3.3 Fragility of buildings before retrofitting

The fragility curve for each building before and after retrofitting was plotted with probability of exceedance on the y-axis and PGA values on the x-axis. The fragility curves before and after retrofitting are shown in Figure 6. Also, the mean fragility curve was drawn to present the average seismic vulnerability of the selected buildings and is presented in Figure 7. The fragility curves obtained for each building are of different natures and characteristics, showcasing the effect of variation in plan, height, and geometric orientation. A vertical line through 0.35g is drawn in order to assess the fragility of the building. The reason for observing 0.35g is that it represents the peak ground acceleration (PGA) for a 475-year return period, which is the Design Basis Earthquake (DBE) of Kathmandu Valley. The point where the vertical line drawn from 0.35g meets the Life Safety (LS) curve gives the probability of exceedance of Life Safety Limit State in the corresponding y-coordinate of that point, which is then used to determine the fragility of the buildings. The reason for observing the Life Safety curve is that Life Safety is defined in the Nepalese seismic design code as the fundamental requirement of structures designed and built in seismic regions of Nepal [24].

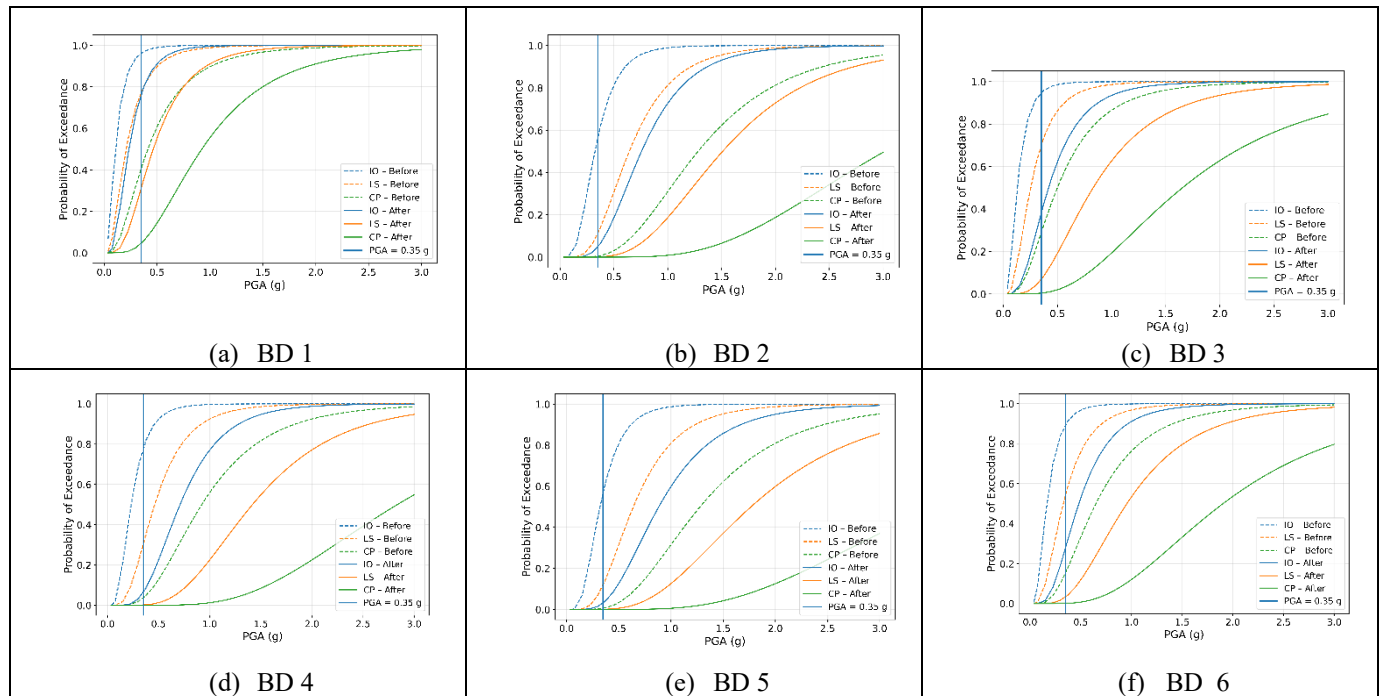
Based on the obtained probabilities, the buildings are classified as Highly Vulnerable, Moderately Vulnerable, and Acceptable Class. The building with the least probability of exceeding the LS limit state at 0.35g is BD 7, whose

exceedance probability is only 6%, followed by BD 2, BD 5, and BD 8, whose probabilities are 11%, 12%, and 12.4%, respectively, and are in the acceptable vulnerability class. The buildings with highest probability of exceedance of LS damage state at 0.35g is BD 1 with 77.5%, followed by BD 3, BD 12, BD 9 and BD 6 whose exceedance probability are 70.2%, 67%, 59% and 54.6% respectively are considered as highly Vulnerable buildings and the buildings like BD 10, BD 4 and BD 11 which have the moderate exceedance probabilities of 42.1%, 30.4% and 20.3% are classified as moderately vulnerable. The probability of exceedance values of three damage states for all 12 buildings before retrofitting is given in Table 6.

It is observed from the above result that the buildings with a greater number of storeys are more vulnerable with a higher probability of exceedance of DBE at the life safety limit state than the buildings with a smaller number of storeys. Consequently, all the buildings that are 5 storeys are found to be highly vulnerable.

3.4 Fragility of buildings after retrofitting

After retrofitting, the performance of buildings is significantly improved. For most of the vulnerable buildings, like BD 3, BD 6, BD 9, and BD 12, the probability was reduced to 8%, 3%, 2%, and 4%, respectively, whereas for BD 1, whose probability of exceeding LS at 0.35g was 77.5% before retrofitting, the probability was dropped to 32% after retrofitting Table 7. Similarly, for the moderately vulnerable buildings like BD 4, BD 10, and BD 11, the exceeding probability is reduced to less than 1% after applying the strengthening solution. Also, for the building which were under the acceptable class, where we conducted the retrofitting to make them code adherent, like BD 2, BD 5, BD 7, BD 8 the exceeding probability was less than 1% which shows that except for building BD 1 the retrofitting technique adopted in this study has worked well in reducing the vulnerability and increasing the performance of the residential RC buildings.



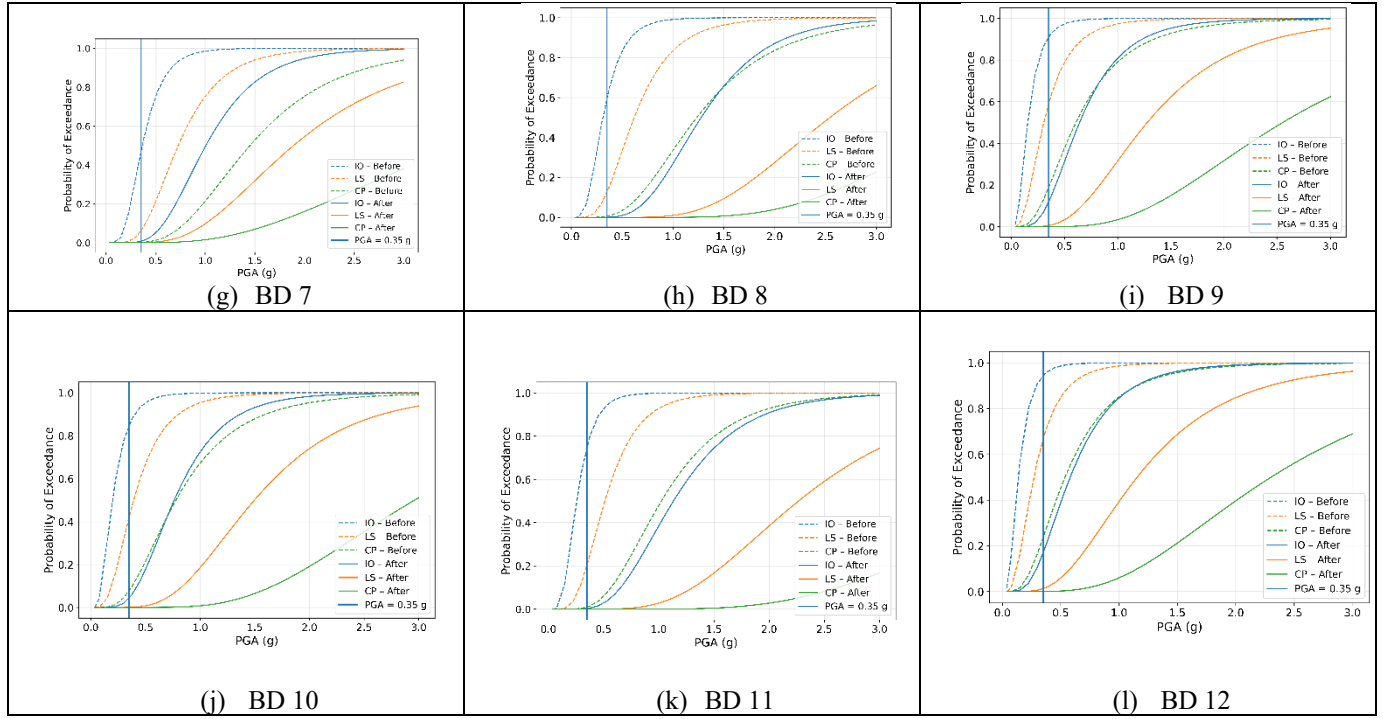


Fig.5 Fragility Curves Before and After Retrofitting for all building models

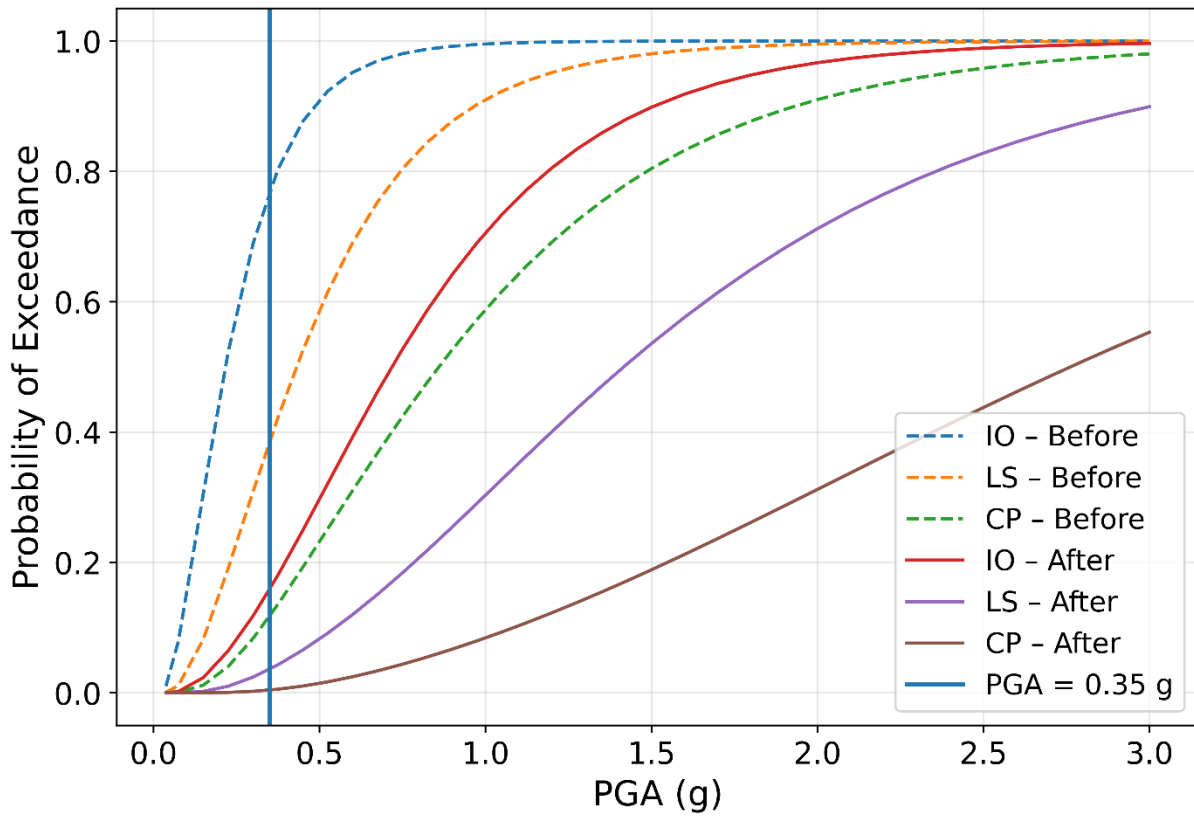


Fig.6 Mean Fragility Curves Before and After Retrofitting

Table 6: Probability of exceedance (%) for 0.35 PGA for buildings before retrofitting in percentage

Damage State	BD1	BD2	BD3	BD4	BD5	BD6	BD7	BD8	BD9	BD10	BD11	BD12	Mean
IO	95.5	56	94.4	77.3	56.4	89.08	46	59.05	90.7	84.3	74.3	94.16	77
LS	77.5	11	70.2	30.4	12	54.6	6	12.4	59	42.1	20.3	67	39
CP	40.5	<1	28.7	3.7	<1	15.52	<1	<1	18.9	7.9	1.06	23.71	15

Table 7: Probability of exceedance (%) for 0.35 PGA for buildings after retrofitting in percentage

Damage State	BD1	BD2	BD3	BD4	BD5	BD6	BD7	BD8	BD9	BD10	BD11	BD12	Mean
IO	75.7	5	37.9	6.64	3.2	28.04	<1	<1	12.03	4.6	<1	17.17	17
LS	32	<1	8	<1	<1	3	<1	<1	2	<1	<1	4	3.66
CP	4.7	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1

4. Conclusion

This study evaluated the seismic vulnerability of 12 residential reinforced concrete (RC) buildings in the Kathmandu Valley constructed before the 2015 Gorkha Earthquake using Incremental Dynamic Analysis (IDA) and fragility curves developed from 10 ground motion records. The findings from previous research suggest that RC buildings constructed before 2015 in the Kathmandu Valley exhibit considerable seismic vulnerability, highlighting the urgent need for systematic retrofitting interventions. The retrofitting technique examined in this study, RC column and beam jacketing, significantly improved the overall seismic performance of the buildings. Based on the mean fragility curves, the probability of exceeding the LS damage state decreased from about 39% before retrofitting to around 4% after retrofitting, representing a reduction of nearly 90%. Similarly, the exceedance probability of the IO damage state decreased from approximately 77% to 17%, while that of the CP damage state decreased from about 15% to less than 1%, corresponding to reductions of about 78% and 95%, respectively. Another important finding of this study is the remarkable improvement observed in lower-rise buildings, namely BD2, BD3, BD5, BD7, BD8, and BD11, following retrofitting. At a PGA of 0.35 g, the LS exceedance probabilities decreased from 11%, 70.2%, 12%, 6%, 12.4%, and 20.3% to <1%, 8%, <1%, <1%, <1%, and <1%, respectively, representing reductions of approximately 89% to 98%.

Although this study provides valuable insights into the fragility of RC buildings inside Kathmandu valley, certain methodological limitations should be acknowledged. This study used a fast non-linear analysis (FNA) method to perform the non-linear time history analysis due to limitations of time and computational resources. FNA provides an efficient way to approximate nonlinear dynamic response, but it is still a simplified method compared to more rigorous nonlinear time history analysis. This study was carried out using a concentrated plasticity approach where predefined auto-hinges were used to represent material nonlinearity, which may not capture the full range of inelastic behavior compared to the distributed plasticity approach. Furthermore, this study assumed a fixed base condition for all buildings, but, in reality, soil-structure interaction may influence seismic response, particularly in soft soil conditions, and its exclusion represents a limitation of this analysis.

Author contributions

Sudeep Kumar Dahal: Conceptualization, Data Collection, Methodology, Software analysis, Visualization, Investigation, Writing – original draft; **Uttam Shrestha:** Data Collection, Methodology, Software analysis, Visualization, Investigation, Manuscript Review; **Sunil Paudel:** Data Collection, Software Analysis, Investigation, Visualization, Manuscript Drafting and Review; **Kshitij C. Shrestha:** Conceptualization, Visualization, Writing - Review and editing, Supervision.

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Data availability: The dataset generated during and/or analyzed during the current study is available from the corresponding author on reasonable request.

Conflict of Interest: The authors declare no conflict of interest.

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